

# RAM BALAJI S

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[LinkedIn Profile](#)

High-performing RTL Design Engineer with 3 years of professional experience in SoC and FPGA designing. Possessing an eye for detail and familiarity with the EDA tools Synopsys and Cadence as well as the FPGA tool Vivado. Ability to develop and implementation of complex projects. To seek and maintain full-time position that offers professional challenges utilizing interpersonal skills, excellent time management and problem-solving skills.

## Skills

- Verilog, Lint, CDC, Digital Concepts, Debugging
- Perl, Python, Makefile
- Linux, Microsoft Office

## Work History

### PCIe Gen 5

Jun 2020 - Jun 2023

Primesoc Technologies, Coimbatore - In-house

- Understood the functionality of Peripheral component Interconnect Express Generation 5 Layers and with depth in physical layer LTSSM state machine.
- Developed complete physical register module for PCIe Gen5 using Verilog language from scratch and tested with verification team for functionality checks and provided RTL fix for verification failure triage
- Managed the malfunctions in LTSSM state transition and provided RTL corrections according to the protocol
- Carried out the integration process of connecting multiple sub blocks to the top level wrapper file, and managed complete Lint checks for design using Cadence tool and gained strong linting knowledge to make the RTL to synthesizable level.
- Exercised FPGA Validation testing using Xilinx Virtex Ultrascale+ family evaluation board VCU118.
- Have introduction with Zync ultrascale+ family evaluation board ZCU106, and understood basic architectures of FPGA
- Ported and tested the bit file of PCIe Gen3 design and analyzed the data flow behaviors using Integrated Logic Analyzer( ILA ) chip scope signals.
- Strong understanding in Vivado tool flow of synthesis, PnR and bit file implementation along with debug cores.
- Analyzed Timing failures in the design and figured out the failure path by reporting timing in Xilinx Vivado tool, and provided fixes for the same.

## Camera Serial Interface - 2

Oct 2021 - Sep 2022

Marvell Semiconductors, Bangalore - Client

- Grasped MIPI - Camera Serial Interface -2 Protocol and Xplorer SoC Architecture
- Constructed the IP from Synopsys CoreConsultant tool and Integrated the IP's to subsystem level through instantiation using Verilog language.
- Investigated linting flaws in RTL design and fixed all the failings and made the RTL to synthesizable and black boxed all third-party intellectual properties.
- Performed data flow checks, interacted with verification issues, and provided RTL fixes for the verification failures.
- Implemented Tie-off registers for reserved ports and assigned addresses and default values for the registers and generated RAL model using standard scripts.
- Implemented memories from standard scripts and incorporated those memories within the design.
- Synchronization of all clocks and reset used inside the design using standard sync modules and gained knowledge of CDC methods.

## Education

Bachelor of Engineering

Jun 2016 - May 2020

Akshaya College of Engineering and Technology, Coimbatore